INTERACTIVE END-OF-CHAPTER EXERCISES

CDMA - ADVANCED

This time, assume there are 2 senders whom interfere with eachother and that the interfering transmitted bit signals are additive. The value received at a receiver, however, is now the sum of the transmitted bits from all senders.

Assume that sender 1 has a CDMA code of (-1, -1, 1, -1, -1, 1, 1, -1, -1) and sender 2 has a CDMA code of (-1, 1, -1, -1, -1, 1, -1, 1, -1) and their combined output is: (2, 0, 0, 2, 2, -2, 0, 0, 2)

QUESTION LIST

1. Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

2. Assuming you are receiver 2, what is the decoded bit? If it can't be done, answer with 'n/a'

SOLUTION

1. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

2. Because the CDMA codes weren't well chosen, interference will render the receiver unable to decode the original bit

QUESTION 1 OF 2

Assuming you are receiver 1, what is the decoded bit? If it can't be done, answer with 'n/a'

Answer

CHECK HINT SKIP

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